Phobos Exploration Propulsion System

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***Abstract*—This report presents a propulsion system designed for a mission to Phobos, the largest and closest of the Martian moons, emphasizing critical trade-offs between electrical and chemical propulsion, propellant choices, and thruster configurations. The report offers practical solutions by providing Commercial Off-The-Shelf (COTS) options for each essential component. The study concludes by developing a comprehensive Pipes and Instrumentation Diagram (P&ID) centered around the selected thrusters. Furthermore, MATLAB simulations are employed to model the system's behavior, specifically tracking changes in pressure and temperature. This holistic approach contributes to advancing the feasibility and practical implementation of an efficient propulsion system tailored for future space exploration missions to Phobos.**

***Index Terms*—propulsion, Phobos, thruster, trade-off, design, modeling, propellants, Mars**

I. Introduction

This work aims to design the propulsion system for a spacecraft with a scientific mission to explore Phobos, which will be launched from Kourou on the 2nd of December of 2028, in a dedicated Ariane 62 launcher.

It will be deployed on an Earth escape orbit with $v_\infty = 2,5$ m/s, which sets the maximum launch mass to 2,6 tons [13]. Further mission-wide requirements are the minimum mission time of 2 years while orbiting Phobos, and the need for de-orbiting into the Martian atmosphere at the End Of Life (EOL). The Δv required for each maneuver and the mass of all the other spacecraft subsystems was taken from the provided initial parameters.

Specific requirements and assumptions:

1) The propulsion system shall provide the necessary thrust and ΔV, required to perform the foreseen mission manoeuvers (Table I);
2) The propulsion system shall provide torques to compensate the main engine misalignments and for all other attitude and orbit control maneuvers, including safe mode operation and reaction wheel offloading, as required by mission planning;
3) The subsystem shall have inhibits against critical failures leading to personnel hazard (e.g. while handling the spacecraft);
4) The propulsion system shall include single failure-tolerant measurement of the propellant pressure;
5) The propulsion system shall provide means to isolate potential external leakages and/or failures of thruster flow control valves;
6) The propulsion system shall include access points (valves) for filling and draining the propulsion system on the ground;
7) The time for the Mars Orbit Insertion (MOI) maneuver must not exceed 30 minutes;
8) The spacecraft shall weigh less than the maximum launch mass.

In the following chapters detailed trade-off analyses are performed and sequentially used to design an optimized propulsion system for this specific mission.

II. Methodology

This study will cover all aspects of the propulsion system's preliminary design, from the mission definition to the final mass budget and system scheme.

Project Phases:

- Mission Definition, Requirements and Constraints;
- Propulsion System Concepts;
- Trade-Offs/Concept Evaluation;
- Tank Modeling and Physical Properties Analysis;
- P&ID Analysis;
- List of Equipment;
- Final Mass Budget.

As explained deeper in section IV and shown in Figure 1, three different trade-offs were performed to evaluate the possible concepts. Initially the feasibility of using electrical propulsion is evaluated. Subsequently, the first trade-off compares conventional propellants commonly used today with potential green alternatives. Finally, the second trade-off evaluates the performance of different COTS thruster/main engine configurations. Considering their results, the optimal propulsion system for this mission was designed, using available components in the market.

After settling on an appropriate configuration, a deeper propulsion system analysis has been conducted, including tank modeling, P&ID, and a complete equipment list.

The mass budget was computed using a self-developed Matlab code, which was used iteratively in all project phases.

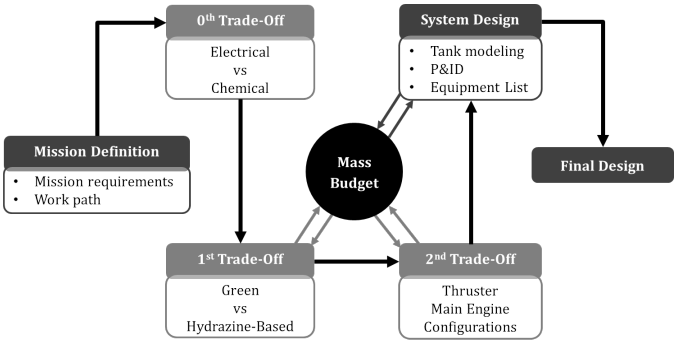


Fig. 1. Work structure.

III. Overall Spacecraft

Table I shows the distribution of Δv for each mission phase.

TABLE I
REQUIRED MISSION MANEUVERS AND THEIR RESPECTIVE ΔV CONSIDERING A MEDIUM-TO-HIGH THRUST PROPULSION SYSTEM.

Maneuver	Δv [m/s]
1) Trans-Mars Injection	590
2) Trajectory Correction Manoeuvres	85
3) Mars Orbit Insertion	835
4) Inclination Change Manoeuvre	95
5) Phasing Orbit Insertion	760
6) QSO Insertion	20
7) Orbit Maintenance	100
8) De-Orbit Manoeuvre	560
Total	3045

The masses of all the subsystems independent from the propulsion system are given in Table II.

TABLE II
MASSES OF ALL THE SPACECRAFT SUBSYSTEMS.

Subsystem	Mass [kg]
Science Instruments (INS)	50
AOGNC	40
Communication (COM)	25
Data Handling (DH)	15
Mechanisms (MEC)	55
Thermal Control (TC)	15
Deployable Surface Probe	20
Launch Vehicle Adapter	85

The made assumptions and used margins and empirical formulas are outlined in Table III. These formed the basis for the mass budget iteration process.

The spacecraft's propulsion system consists of one or more main engines for large Δv maneuvers and thrusters for attitude and orbit control (AOC) maneuvers. The propellant feed and pressurization system were designed according to the chosen configuration.

IV. Propulsion System Design

This chapter provides a detailed description of each trade-off analysis, including their rationale and the outcomes obtained from each. The result of the preceding trade-off analysis is used as the input for the subsequent one. The second trade-off analysis elucidates the process of selecting the optimal configuration for the mission. Following this, a detailed development of the system encompassing the thrusters/engines is undertaken, culminating in the presentation of results.

TABLE III
EMPIRICAL RELATIONS FOR MASSES.

Topic	Assumption
Structural Mass	$m_{STR} = 0.0224 \cdot m_{wet} + 27.823$ kg
Power System Mass	$m_{PWR} = 0.0758 \cdot P_{el,req} + 31.741$ W [1]
Harness Mass	10% of S/C dry mass without system margin
System Margin	20% of dry mass
Ullage Volume	2% of propellant volume

[1] All subsystems except propulsion require a total of 550 W.

A. Zeroth Trade-Off Analysis

An initial study was conducted to explore the potential use of electrical propulsion for the spacecraft's main engine and/or AOC thrusters.

1) Main engine: The greatest challenge is satisfying requirement 7, which states that the MOI cannot exceed 30 minutes. This has a strong implication on the required thrust level.

Electrical thrusters such as the Ariane Group RIT 2X radiofrequency ion thruster [12] and the Aerojet Rocketdyne AEPS hall thruster [14] were considered for this study. In all cases, the time required to perform the MOI burn surpassed the 30-minute requirement. Additionally, their minimum lifetime is inferior to the total expected mission duration when using electrical propulsion for the main engine [41].

2) Thrusters: Various COTS electrical propulsion thrusters were studied and their suitability as an AOC system was evaluated. Within the results of this study were thrusters such as the Enpulsion Micro R^3 [9] and the Ariane Group RIT 10 Evo [12], which are designed for microsatellites to serve as their main thruster. These options were discarded as they are developed for long-duration burns instead of the short ones required for attitude control, which will be the main function of the thrusters, as stated in requirement 2. Furthermore, their weight is generally larger than a chemical thruster, as they require additional hardware such as a Xenon tank, power processing unit, and flow control units. The extra required hardware adds weight, volume, complexity, and risk, making this option unattractive when compared to chemical thrusters.

This analysis concludes that employing electrical propulsion, whether for the main engine or AOC, does not align with the requirements of this mission. Therefore, chemical propulsion will be analyzed in the next trade-off analysis.

B. First Trade-Off Analysis

This trade-off aims to evaluate two types of propellants: conventional hydrazine-based and green propellants. The criteria for comparison were established based on the studies on propellant selection methods proposed by V.Bombelli et al.

[39] and O. Frota et al. [8]. Only the most relevant criteria were considered, listed in Table XII. Weights are introduced to account for the relevance of the different aspects.

This work has not considered dual-mode propulsion systems due to their design complexity compared to single-mode bipropellant systems, especially concerning the prediction of residual oxidizer/fuel after each maneuver. Moreover, dual-mode thrusters operating with monopropellants perform considerably worse than with bipropellants.

Three different bipropellants were selected as possible candidates, listed in Table IV. MMH/NTO was chosen to represent hydrazine-based propellants because of their frequent use in space propulsion applications. The choice of propellants representing green propulsion is based on the most promising thrusters currently in the market.

1) Input data: A summary of all specific input data for each configuration and the respective references can be found in Table IV. Since no readily available main engine uses Ethanol/H2O2 or Propylene/N2O, the input data was derived from research papers and modeling software RPA. For this trade-off, a speculative tank model of 50 kg and 1,5 m^3 was used for both propellant and oxidizer in every proposed configuration. As an initial rough approximation, the Return Module of the MMX mission [16] was taken as a reference for the number of main engines (4) and thrusters (20) used to fulfill the mission. Furthermore, the following simplification was made to allocate each Δv maneuver to either the main engines or the thrusters. In particular, referring to Table I, maneuvers 1, 3, 4, 5, and 8 are only performed by the main engines and maneuvers 2, 6, and 7 by the thrusters.

TABLE IV
TYPES OF PROPELLANT - MAIN ENGINE & THRUSTER

	MMH & NTO	Ethanol & 90% H_2O_2	Propylene & N_2O
ρ_{ox} [kg/m^3]	1448 [5]	1390[1] [31]	1220[2] [27]
ρ_{fu} [kg/m^3]	878 [5]	789 [5]	611[2] [5]

[1]At 20°C. [2]At boiling temperature.

		MMH & NTO	Ethanol & H_2O_2	Propylene & N_2O
Main Engine	Model & Company	S400-15, Ariane G. [3]	[40]	-[3]
	ISP_{vac} [s]	321	311,4[1]	278,6
	O/F	1,65	4,5	8,85
	Thrust [N]	425	482	450
	Mass [kg]	4,30	4,30[2]	4,30[2]
Thruster	Model & Company	R-6D, Aerojet [30]	Benchmark [33]	B20, Dawn AS. [1]
	ISP_{vac} [s]	294	310	285
	O/F	1,65[2]	4,5[4]	8,85 [6]
	Thrust [N]	22	22	16,7
	Mass [kg]	0,45	0,60[5]	0,60

[1]Recalculated based on higher expansion ratio (330) and 0,9 efficiency; [2]Assumed same as S400-15; [3]Results from RPA; [4]Assumed same as main engine; [5]Assumed same mass as B20 thruster.

2) Criteria Definitions: The considered criteria are listed in Table XII and detailed in this chapter.

a) Propulsion System Total Mass: The first important criterion in selecting the best propellant is the mass budget. This was calculated with a self-developed MATLAB code considering the data in Table II and relations in Table III. Figure 2 shows the mass output.

b) Propellant Volume: The propellant volume budget criterion is essential for comparing the size of various propulsion systems. A smaller volume results in reduced tank mass. Additionally, having the same volume for oxidizer and fuel facilitates using identical tanks, thereby reducing costs.

TABLE V
CRITERIA AND WEIGHTS [39]

Criteria	Weight [%]
Propulsion System Total Mass	25
Propellant Volume	20
Material Compatibility	12,5
Hazards	10
Storability	12,5
Ignition & Combustion	10
Cost	10
Technological Readiness	20

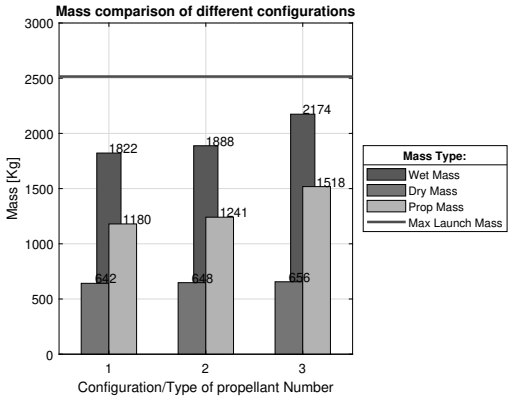


Fig. 2. Wet, dry, and propellant mass of each configuration.

The same code used for mass calculations is used. The output is displayed in Figure 3.

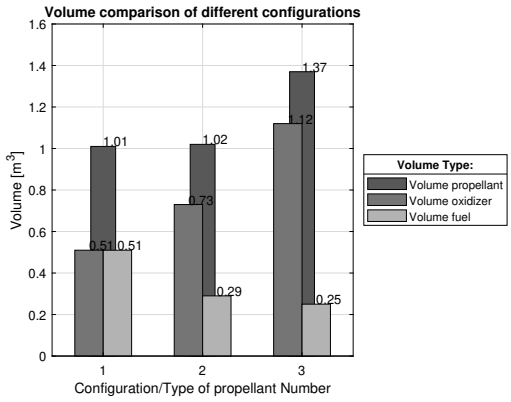


Fig. 3. Propellant volume, oxidizer volume, and fuel volume of each configuration.

c) Material Compatibility: Material compatibility has been analyzed using tables 10 and 12 from [5]. Long-term compatibility, corrosion of materials, chemical poisoning, unwanted material deposition, susceptibility to auto-decomposition, and explosion were all relevant for grading. For each configuration, the compatibility level is determined by the lowest contribution from the fuel and oxidizer. To summarize the results: MMH/NTO (High-

/Medium) → Medium; Ethanol/H_2O_2(High/Low) → Low; Propyle/N_2O(High/Medium) → Medium.

d) Hazards: The hazard evaluation has been conducted using as reference tables 9 and 11 from [5]. Penalty points for each category are assigned according to the following criteria: 2 penalty points for the least favorable, 1 penalty point for moderately favorable, and no penalty for the most favorable. The points are then totaled for both propellants, resulting in the following scores: MMH/NTO -25 points (fuel: -18; ox: -7); Propylene/N_2O -6 points (fuel: -3; ox: -3); Ethanol/H_2O_2 -3 points (fuel: 0; ox: -3).

e) Storability: Storability is critical for space missions, with factors like boiling/freezing point and vapor pressure being key. MMH/NTO is well-established as highly storable, having been used for decades. However, green propellants face challenges in this regard. N_2O offers self-pressurization benefits but requires careful management due to its tight storage window. Hydrogen peroxide needs stabilizers to mitigate decomposition issues, while Propylene requires high pressures for storage. As a conclusion to these properties, points are assigned as follows [5]: Level 1 for MMH/NTO; Level 2 for Ethanol/H_2O_2; Level 3 for Propylene/N_2O.

f) Ignition & Combustion Properties: Combustion in propulsion systems typically requires an igniter system, but some propellants ignite spontaneously upon contact, known as hypergolic propellants [34]. This property reduces system complexity and failure probability in space systems.

Of the discussed propellant combinations, MMH/NTO consistently exhibits pure hypergolic behavior. While H_2O_2 shows promising hypergolicity, it lacks full confirmation [7], [17], [19], [28], [40]. Conversely, evidence of hypergolic behavior for N_2O with Propylene is not publicly available. Respective points in this criteria are given as follows: Level 1 (pure hypergolicity) for MMH/NTO; Level 2 (partial hypergolicity) for Ethanol/H_2O_2; Level 3 (non-hypergolic) for Propylene/N_2O.

g) Costs: Economic considerations are pivotal in selecting rocket propellants, primarily focusing on reducing costs associated with handling and accident prevention. According to [5], total costs are divided into various components, including propellant and raw material costs, hardware for transfer and handling, ground service equipment, personal protection, and residual disposal. These costs are expected to vary between toxic and non-toxic propellants. Toxic propellants like MMH and NTO typically require complex flow systems, extensive cleaning and decontamination procedures, specialized worker training, regular medical monitoring, and the use of personal protective equipment. Therefore, adopting less hazardous propellants can enhance safety, streamline operations, and reduce overall costs.

h) Technological Readiness: The technological readiness of a propulsion system is typically quantified using the Technology Readiness Levels (TRLs) scale. This criterion helps to measure the maturity of a technology based on a range from 1 (Basic Technology Research) to 9 (Systems Test, Launch, and Operations) [26].

With reference to this scale, the family of Hydrazine-based propellants has been used widely in the past decades receiving highest level. Regarding green alternatives, the development is ongoing, and readiness levels are limited due to the low number of relevant missions conducted.

Per NASA, both Ethanol/H2O2 and Propene/N_2O have already been used for some missions (single instance for Ethanol/H2O2) but only for small thrusters [25]. Regarding these reasons, this report designates the technological readiness of MMH/NTO as the highest (level 3), while assigning a middle level (level 2) to the Propylene/N_2O and Ethanol/H_2O_2 configurations.

3) Results - 1st Trade-Off Analysis: According to the weights specified in Table XII, the results are calculated by multiplying each point by its corresponding weight and then summing them for each propellant. These results are depicted in Figure 4 using a spider graph.

Ethanol/H_2O_2 exhibits superior performance in total mass and safety, albeit lacking material compatibility. Propylene/N_2O offers the best cost efficiency, yet suffers from the highest system mass and lowest ignition performance. Notably, both green propellants exhibit limited technological readiness. Conversely, MMH/NTO excels in mass and volume efficiency, ignition, and storability. Due to these factors, hydrazine-based propellants were deemed the optimal choice for the objectives of this mission.

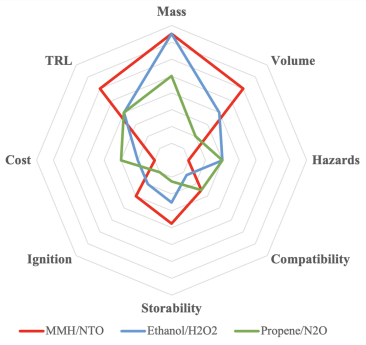


Fig. 4. Spider graph of the first trade-off analysis.

C. Second Trade-Off Analysis

The aim of the second trade-off is to investigate various configurations of thrusters and main engines utilizing hydrazine-based propellants and choose the optimal one for the mission. Unlike previous approaches, this analysis considers a total thrust of ∼1000 N for main engines and ∼500 N for thrusters. These thrust values were determined based on the minimum required thrust level under the 7th mission requirement.

Furthermore, the reliability of the engine must be factored into the equation. Consequently, the minimum number of engines necessary to meet the time constraint was selected based on the thrust provided by each configuration's engine. The need and use of more engines increases risk as the mission relies on the correct functioning of more components to be successful. One could see it the other way around, but that

would only be true if the mission were redundant in the amount of available main engines, which is not the case. In summary, configurations with medium thrust employ two engines, while configurations with high thrust utilize only one engine.

Consistency was ensured using the same bipropellant tank, Ariane Group's OST 25/0 [4], for every configuration listed in Table VI.

TABLE VI
SECOND TRADE-OFF CONFIGURATIONS

	Main engines	Thrusters
	1) Medium Thrust, MMH/NTO	
Model	2 x S400-15 [3]	20 x R-6D [30]
Thrust [N]	2 x 425	20 x 22
	2) High Thrust, MMH/NTO	
Model	1 x R-42 [30]	20 x R-6D [30]
Thrust [N]	1 x 890	20 x 22
	3) Medium Thrust, MMH/MON	
Model	2 x R-4D-15 HiPAT [29]	20 x DST-12 [21]
Thrust [N]	2 x 445	20 x 22
	4) High Thrust, MMH/MON	
Model	1 x LEROS 4 [24]	20 x DST-12 [21]
Thrust [N]	1 x 1000	20 x 22
	5) Medium Thrust, MMH/MON	
Model	2 x LEROS 2b [23]	20 x DST-12 [21]
Thrust [N]	2 x 420	20 x 22
	6) Medium Thrust, Hydrazine/MON	
Model	2 x 450N Thruster [2]	20 x 22N Thruster [2]
Thrust [N]	2 x 450	20 x 22

1) Criteria Definitions: The criteria presented by V. Bombelli et al. [39] and O. Frota et al. [8] were revisited for this analysis. Given the similarity in properties among hydrazine-based propellants, they were updated, as listed in Table VII. Consequently, adjustments in weights were also made.

TABLE VII
CRITERIA AND WEIGHTS [39]

Criteria	**Weight [%]**
Propulsion System Total Mass	25
Propellant Volume	20
Tank Similarity	10
Heritage	25
Reliability	20

a) Propulsion System Total Mass: The calculation was made with a modified version of the previously used Matlab code, now considering the total thrust and the chosen tank model. The results are visualized in Figure 5.

b) Propellant Volume: As previously noted in the preceding trade-off, a smaller volume leads to a decrease in tank mass. The code utilized for mass calculations was also employed for the volume computation. The corresponding output is presented in Figure 6.

c) Tank Similarity: Tank similarity permits quantifying each configuration's volume difference between propellant and oxidizer. The same tank volume leads to cost benefits and easier implementation.

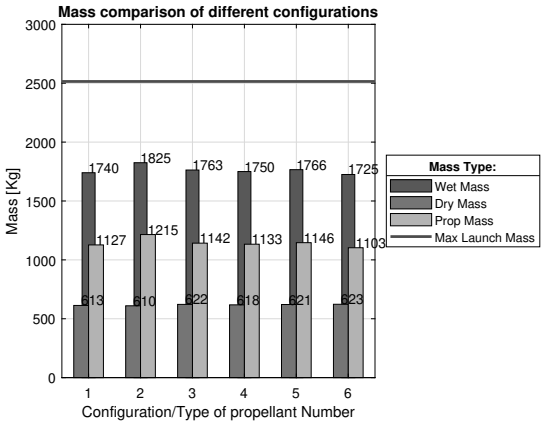


Fig. 5. Wet, dry, and propellant mass of each configuration.

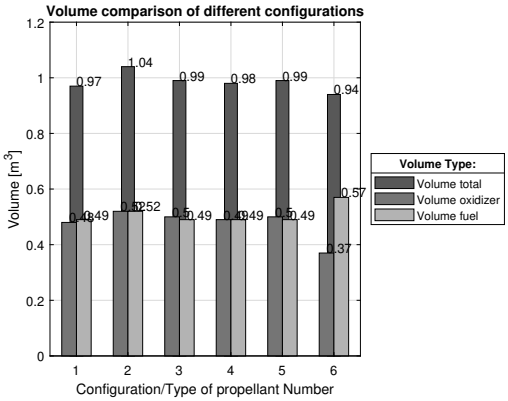


Fig. 6. Total, oxidizer, and fuel volume of each configuration.

d) Heritage: As done in the first trade-off, the heritage refers to the Technology Readiness Levels (TRLs) scale. This criterion helps to measure the maturity of a technology [26].

e) Reliability: The reliability criterion deals with the number of main engines used. Employing two main engines introduces an additional point of failure in the propulsion system, consequently elevating the risk of mission failure.

2) Results: A series of calculations were conducted after establishing the weights for the five criteria to identify the optimal configuration for the mission, Table VII. These results can be seen in Figure 7.

Mass and volume performance exhibit minimal variation across all configurations, with significant disparities observed in other criteria. Particularly, the 6th configuration displays inferior reliability and tank similarity performance, while the 4th configuration fares poorly regarding heritage. Considering these factors, the 2nd one emerges as the configuration that consistently performs well across all criteria, closely followed by the 4th. Consequently, the 2nd configuration has been selected as the optimal choice for the propulsion system.

As the result of all trade-off analysis chemical propulsion for both main and AOC thrusters, using MMH/NTO was chosen. The optimal market option was established to be the following models for the main engines and thrusters respec-

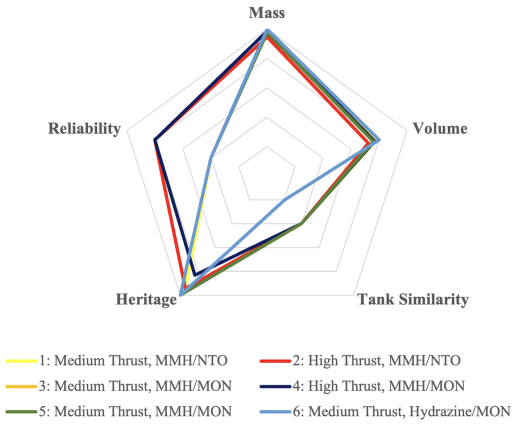


Fig. 7. Spider graph of the second trade-off analysis.

tively: Aerojet Rocketdyne R-42 and Aerojet Rocketdyne R-6D.

D. Propellant and Pressurant Tank Modeling

Once the optimal configuration was selected, the propulsion system was studied more thoroughly. This section aims to compare different alternatives in terms of tanks available on the market and obtain at the end a physical characterization of the propulsion system in terms of pressure, temperature, volume, and mass behavior throughout the mission. To achieve a higher level of detail in the system, the following measures were implemented on the studied configuration:

- While the MMX reference remains relevant, further analysis has prompted consideration of the propulsion module instead of the return module [16]. Consequently, this involves utilizing 4 thrusters with an additional 4 for redundancy, resulting in a total of 8 thrusters for the propulsion system;
- Out of 8 thrusters, 4 of them are fired during phases 2, 6, and 7 while 2 of them are used together with the main engine during the other phases to account for torque imbalance caused by the main engine (see Table I);
- Fuel and oxidizer tanks are differentiated and considered in the right quantity;
- Pressurant tank (Helium) is introduced in the tank modeling and mass budget analyses.

Table VIII presents the possible tank alternatives analysed. Important aspects to consider are:

- At this point a similar MMH and NTO volume of around 600 L is needed as well as around 100 L for helium (depending on the operating pressure) from the mass budget calculation;
- To prevent any potential propellant backflow, a minimum pressure difference of 50 bar between the pressurant and MMH/NTO tanks was targeted at the end of the mission.
- Considering a chamber pressure within 15-20 bar, taking as assumption line losses of 2 bar, and 10% margin on the MEOP, the lowest possible MEOP value for propellant tanks is 18,7 bar;
- Especially for pressurant tank, higher pressure (within MEOP), means less tank volume needed and therefore, less mass, but heavier components overall;
- With the previous points in mind, the best configuration has the lowest number of tanks. Moreover, higher volume leads to lower required pressure (for the Helium tank) although a weight increase must be considered.

The tank evaluation and the calculation of physical properties were implemented using a self-developed MATLAB code for this specific purpose.

TABLE VIII
EVALUATION OF MMH/NTO TANKS.

Model	Volume [L]	Mass [Kg]	MEOP[1] [bar]
Fuel/Oxidizer Tanks			
OST 25/0 [4]	331	21	20
OST 22/X [4]	700	36	19.5
OST 23/0 [4]	769	31.7	17.5
650L [18]	650	31	16.5
769L [18]	780	37	17.5
Pressurant Tanks			
PVG 40-75 [22]	40-75	8.5-14.5	310
PVG 80-120 [22]	80-120	23.5	310

[1] Maximum Expected Operating Pressure.

TABLE IX
POSSIBLE TANK CONFIGURATIONS FOR MMH/NTO/HELIUM.

Models	Tank #	Tot. V. [L]	Tot. M. [kg]	He P. [bar]
1) OST 25/0 - PVG 75	2 / 2 / 2	662 / 662 / 150	113	247
2) OST 25/0 - PVG 120	2 / 2 / 1	662 / 662 / 120	107,5	285
3) OST 22/X - PVG 75	1 / 1 / 2	700 / 700 / 150	101	237
4) OST 22/X - PVG 120	1 / 1 / 1	700 / 700 / 120	95,5	256
5) OST 23/0 - PVG 120	1 / 1 / 1	769 / 769 / 120	86,9	247
6) 769L - PVG 120	1 / 1 / 1	780 / 780 / 120	97,5	256

The fourth configuration offers the most favorable balance between tank mass and helium tank pressure while meeting the minimum pressure requirements for MMH and NTO tanks. Consequently, this tank combination has been selected for the propulsion system.

It is imperative to highlight that the selected Helium tank model, PVG-120, manifests a nominal operational threshold of -30°C. In adherence to this stipulation, albeit Helium is inherently resilient to even lower subzero temperatures, the pressurant pressure has been augmented to 276 bar. This adjustment facilitates the attainment of an elevated minimum temperature within the Helium tank. Subsequently, the pressurant pressure at the denouement of the entire mission stands at 97 bar.

Regarding the temperature profiles, Helium and the propellant contained within the MMH and NTO tanks are analyzed, assuming instantaneous equilibrium temperature achievement subsequent to the infusion of pressurant into the propellant tanks.

Ultimately, a comprehensive equilibrium temperature of 20°C for the whole spacecraft throughout the duration of the mission has been postulated.

The physical characterization of the propulsion system, including pressure and temperature behavior throughout the mission, is depicted in Figure 8.

E. Piping and Instrumentation Diagram

For the P&ID as seen in Figure 9, the requirements 3, 4, 5, and 6 are relevant. These requirements are addressed using solenoid latching valves upstream of the thrusters and main engine. These increase the ground crew's safety and prevent propellant leakage during the mission. With the use of fill and drain valves and test ports, testing the system components is made possible. Single failure tolerance is ensured by building redundancies in the valves and sensors.

F. List of Equipment

Table XI shows all the propulsion system equipment. A 5% margin was considered for all the items whose mass is provided by its manufacturer and 20% for all the items whose mass was assumed.

G. Mass Budget Update

As the project's final step, the mass of the total propulsion system was computed. The resulted mass from Table XI has been added to the mass budget.

A summary of the propulsion system masses designed is shown in Table X. Moreover, 602 L and 601 L (respectively MMH and NTO) are necessary in propellant tanks at the beginning of the mission and the time requirement on the Mars Orbit Insertion maneuver is successfully fulfilled in 22 minutes.

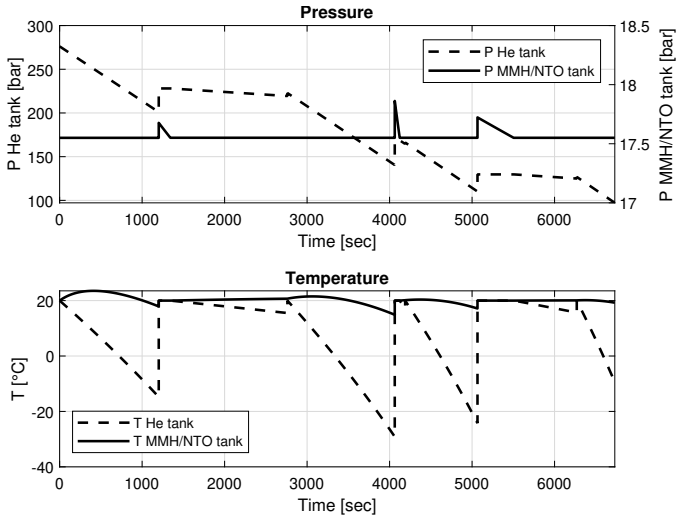


Fig. 8. Propulsion system pressure and temperature behavior.[1]
[1]MMH and NTO curves have been depicted here as coincident, although they are not, as otherwise the difference would not be noticeable.

TABLE X
SUMMARY OF SPACECRAFT MASS.

At Launch	Value [kg]	At EOL	Value [kg]
Mass MMH	529	Mass MMH	25
Mass NTO	870	Mass NTO	41
Mass He[1]	6	Mass He[2]	6
Dry Mass	665	Dry Mass	665
Wet Mass	2070	Wet Mass	737
Total Launch Mass[3]	2155		

[1] Bigger quantity in pressurant tank; [2] Bigger quantities in MHH and NTO tanks; [3] Launch vehicle adapter included.

V. CONCLUSION AND OUTLOOK

A three-step trade-off approach has been employed to ensure optimal design decisions as part of the propulsion system design process. Despite the emergence of green propellants as potential alternatives to hydrazine-based propellants, their low technological readiness level and limited availability of main engine options have hindered their utilization for this mission. However, with an increasing number of demonstrations in orbit and proof of concept, it is anticipated that their use is likely to become more prevalent in the space propulsion sector.

The trade-off analysis converged into an MMH/NTO-propelled system with COTS Aerojet Rocketdyne thrusters and main engine. With this arrangement, all requirements are met and an additional 445 kg of launch mass can be used for unplanned weight changes, emergencies, or additional scientific payload. Therefore, the mission's feasibility has been proved from a propulsion system point of view, and this work may serve as a guide for the preliminary design of similar destination missions.

In the next stages of development, control authority must be calculated to ensure precise spacecraft maneuverability. A CAD model of the propulsion system needs to be created to visualize and refine the system's design. Empirical formulas should be replaced by more precise structure and power subsystem mass estimations. Cost estimation is required for financial planning purposes. System components must undergo testing to ensure functionality over the duration of the mission. Equipment has to be purchased from suppliers to meet project requirements. The integration and testing of components need to be carefully executed to ensure seamless operation. Both component and system tests must be conducted to validate performance and functionality and to confirm mission readiness.

VI. TEAM SETUP

Ali B. Akbey: Creation of pathway for trade-off analyses and choice of configurations, literature review, P&ID.

Jessie: List of equipment, P&ID.

Leonardo Eitner: Feasibility analysis of electrical propulsion, list of equipment.

Matteo Solesio: Mass Budget Calculation, Tank Modeling.

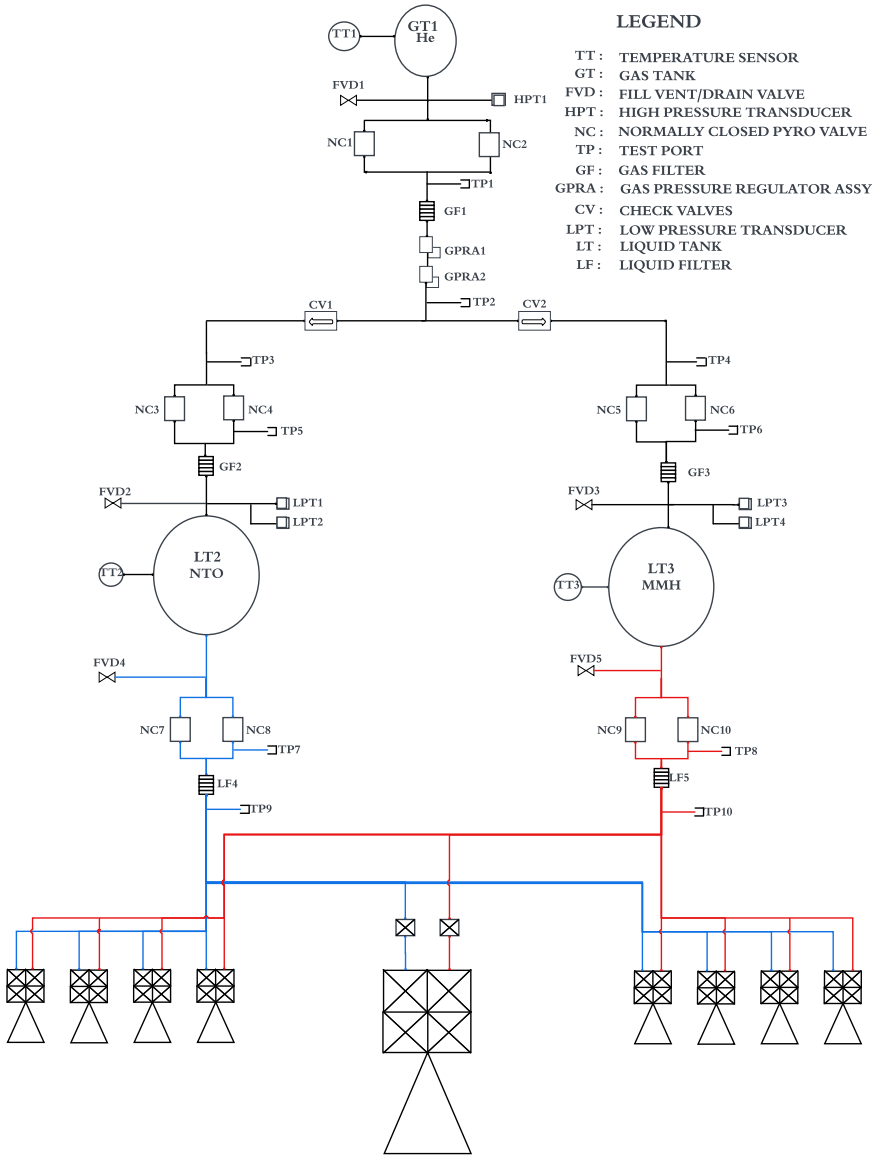


Fig. 9. Piping and Instrumentation Diagram

TABLE XI
LIST OF PROPULSION SYSTEM EQUIPMENT.

Description/Item	Type	Amount	Mass [kg]	Total mass [kg]	Total power [W]	Margin [%]	Mass w/ margin
T. Transducer	Aerospace T. Probes Series 44C [15]	3	0,1	0,3	0	20	0,36
Fill/Drain Valve	Fill & Drain Valves [10]	5	0,09	0,45	0	5	0,473
H. P. Transducer	MINI P. TRANSDUCER [32]	1	0,125	0,125	1 x 0,3	5	0,131
L. P. Transducer	MINI P. TRANSDUCER [32]	4	0,125	0,5	4 x 0,3	5	0,525
Pyrovalve	Pyrotechnic Valve [11]	10	0,16	1,6	0	5	1,68
Liquid Filter	¼” F1D10638-01 [38]	2	0,18	0,36	0	5	0,378
Gas Filter	¼” F1D10588-01 [37]	3	0,024	0,072	0	5	0,076
Check Valve	¼” V0E10495-01 [35]	2	0,136	0,272	0	5	0,286
P. Regulator	50X1463 Regulator [20]	2	1,8	3,6	0	5	3,78
Test Ports*	Fill & Drain Valves [10]	10	0,09	0,9	0	5	0,945
Latching Valves	¼” V1E10405-04 [36]	2	0,34	0,68	0	5	0,714
Pipe Lines	Guess	-	5	5	0	20	6
Propellant Tank	Bi. Tank. Model OST 22/X [4]	2	36	72	0	5	75,6
Pressurizer Tank	PVG Family 80-120 [22]	1	23,5	23,5	0	5	24,675
Thrusters	R-6D 5-lbf (22N) [30]	8	0,454	3,632	8 x 5	5	3,814
Main Engine	R-42 890N (200 lbf) [30]	1	4,53	4,53	1 x 46	5	4,756
Total	-	57	-	118	87,5	-	124

VII. Personal Contribution - Ali B. Akbey

My contribution to the work included the following:

1) Coordinating and overseeing various aspects of the team's activities
2) Initial steps of development in mass budget calculation code
3) Establishing the 1th and 2nd trade-off processes:
   a) 1th Trade-Off
      i) Literature review: Comparison of green and hydrazine based propellants
      ii) Selection of green propellants for consideration
      iii) Identifying COTS engines/thrusters
      iv) Rough preliminary design of main engines for H_2O_2/Ethanol and N_2O/Propylene
      v) Literature Review: storability, ignition&combustion properties
   b) 2nd Trade-Off
      i) Modification of criteria&weights
      ii) Selecting COTS main engine and thrusters and creation of configurations for evaluation
4) Initial steps of development in tank modeling code
5) Development of piping&instrumentation diagram

A. Mass Budget Code

The need to develop a code capable of calculating the required propellant mass after each thrust maneuver was crucial to compare each configuration later evaluated in trade-off processes. I contributed to the development by helping to implement relevant equations in an iterative framework.

B. First Trade-Off Analysis

This trade-off aims to evaluate two types of propellants: conventional hydrazine-based and green propellants. The criteria for comparison were established based on the studies on propellant selection methods proposed by V. Bombelli et al. [39] and O. Frota et al. [8]. Only the most relevant criteria were considered, listed in Table XII. Weights are introduced to account for the relevance of the different criteria.

Three different bipropellants were selected as possible candidates: MMH/NTO, Ethanol/H_2O_2 and Propylene/N_2O.

TABLE XII
Criteria and Weights 1th Trade-Off

Criteria	Weight [%]
Propulsion System Total Mass	25
Propulsion System Volume	20
Material Compatibility	12,5
Hazards	10
Storability	12,5
Ignition & Combustion	10
Cost	10
Technological Readiness	20

1) Selection of Thruster&Main Engines:

a) MMH/NTO: ENGINE: S400-15 [3] engine from Ariane Group was chosen to represent hydrazine-based propellants.

THRUSTER: The R-6D model [30] from Aerojet Rocketdyne was selected as the AOC thruster. Although the exact mixture ratio was not specified in the Aerojet data sheet, it was assumed that the MMH/NTO thruster and engine operate at the same mixture ratio.

b) Ethanol/H_2O_2: ENGINE: The design parameters for the main engine were derived from W.P.W. Wieling et al. [40]. The expansion ratio was adjusted to 330, based on S400-15. Subsequently, the vacuum Isp at this expansion ratio was calculated and then multiplied by a determined manufacturing efficiency of 0,9, resulting in an Isp of 311,4 s. The engine mass was assumed to be the same as S400-15.

THRUSTER: The COTS thruster by Benchmark Space Systems [33] is considered. Since Benchmark does not disclose the type of fuel used, it is assumed that ethanol is utilized, given its popularity as an organic fuel and comparable performance of the thruster to the theoretical Ethanol/H_2O_2 performance. Additionally, it is assumed that the O/F ratio of the thruster and main engine is the same. The company datasheet does not provide the thruster's mass. Therefore, it is assumed that Ethanol/H_2O_2 thruster has the same mass as B20 thruster of Dawn Aerospace(further detailed under Propylene/N_2O section). .

c) Propylene/N_2O: ENGINE: Due to a lack of COTS main engines utilizing these propellants, the Rocket Propulsion Analysis software was employed to gather rough performance data. The input parameters for the software included a chamber pressure of 15 bar based on S400-15, a thrust value of 450 N (averaged from two other configurations). Utilizing N_2O(L) at $184,67K$ as the oxidizer, and C_3H_6(L) as fuel, at an expansion ratio of 330, consistent with S400-15.

The outputs yielded a vacuum Isp of 278,6 s, accounting for a manufacturing efficiency of 0,9. Furthermore, the engine mass is once again assumed to be the same as S400-15.

THRUSTER: The B20 COTS thruster by Dawn Aerospace [1] was chosen for consideration.
The resulting configurations for the 1th trade-off can be seen in Table XIII and Table XIV.

TABLE XIII
Main Engines 1th Trade-Off

	MMH & NTO	Ethanol & H_2O_2	Propylene & N_2O
Model & Company	S400-15, Ariane G. [3]	-	-
ISP_{vac} [s]	321	311,4	278,6
O/F	1,65	4,5	8,85
Thrust [N]	425	482	450
Mass [kg]	4,30	4,30	4,30

TABLE XIV
Thrusters 1th Trade-Off

	MMH & NTO	Ethanol & H_2O_2	Propylene & N_2O
Model & Company	R-6D, Aerojet [30]	Benchmark [33]	B20, Dawn AS. [1]
ISP_{vac} [s]	294	310	285
O/F	1,65	4,5	8,85
Thrust [N]	22	22	16,7
Mass [kg]	0,45	0,60	0,60

2) Criteria Evaluation:

a) Storability: Storability is critical for space missions, with factors like boiling/freezing point and vapor pressure being key. MMH/NTO is well-established as highly storable, having been used for decades. However, green propellants face challenges in this regard. N_2O offers self-pressurization benefits but requires careful management due to its tight storage window. Hydrogen peroxide needs stabilizers to mitigate decomposition issues, while Propylene requires high pressures for storage. As a conclusion to these properties, points are assigned as follows [5]:

- MMH/NTO: Level 1;
- Ethanol/H2O2: Level 2;
- Propylene/N_2O: Level 3.

b) Ignition & Combustion Properties: Combustion in propulsion systems typically requires an igniter system, but some propellants ignite spontaneously upon contact, known as hypergolic propellants [34]. This property reduces system complexity and failure probability in space systems.

Of the discussed propellant combinations, MMH/NTO consistently exhibits pure hypergolic behavior. While hydrogen peroxide shows promising hypergolicity, it lacks full confirmation [7], [17], [19], [28], [40]. Conversely, evidence of hypergolic behavior for N_2O with Propylene is not publicly available and B20 thrusters utilize a spark-based igniter. Respective points in this criteria are given as follows:

- MMH/NTO: Level 1 (pure hypergolicity);
- Ethanol/H2O2: Level 2 (partial hypergolicity);
- Propylene/N_2O: Level 3 (non-hypergolic).

C. Second Trade-off

The configurations in Table XV were chosen COTS as part of the 2nd trade-off analysis. Subsequently, they were thoroughly compared to determine the optimal choice based on the considered criteria.

TABLE XV
CONFIGURATIONS 2nd TRADE-OFF

	Config. 1: Medium Thrust, MMH/NTO	
Model	S400-15 (Apogee Motor) [3]	R-6D [30]
Thrust [N]	425	22
Isp [s]	321	294
Mixture Ratio	1,65	1,61[1]
Mass [kg]	4,3	0,454
Heritage	Extensive	Flight qualified

[1]Assumed same as model R-6F.

	Config. 2: High Thrust, MMH/NTO	
Model	R-42 [30]	R-6D [30]
Thrust [N]	890	22
Isp [s]	303	294
Mixture Ratio	1,65[2]	1,61[1]
Mass [kg]	4,53	0,454
Heritage	Flight proven, in production	Flight qualified

[1]Assumed same as model R-6F; [2]Assumed same as model R-4D-11.

	Config. 3: Medium Thrust, MMH/MON	
Model	R-4D-15 HiPAT [29]	DST-12 [21]
Thrust [N]	445	22
Isp [s]	320,6	321
Mixture Ratio	1,65	1,61
Mass [kg]	5,44	0,64
Heritage	Flight proven, in production	Extensive

	Config. 4: High Thrust, MMH/MON	
Model	LEROS 4 [24]	DST-12 [21]
Thrust [N]	1000	22
Isp [s]	302	302
Mixture Ratio	1,65	1,61
Mass [kg]	8,41	0,64
Heritage	In qualification testing	Extensive

	Config. 5: Medium Thrust, MMH/MON	
Model	LEROS 2b [23]	DST-12 [21]
Thrust [N]	420	22
Isp [s]	319,5	302
Mixture Ratio	1,65	1,61
Mass [kg]	5	0,64
Heritage	Extensive	Extensive

	Config. 6: Medium Thrust, Hydrazine/MON	
Model	450N Thruster [2]	22N Thruster [2]
Thrust [N]	450	22
Isp [s]	329	295
Mixture Ratio	0,94[1]	0,94[1]
Mass [kg]	5[2]	0,77[3]
Heritage	Flight proven	Extensive

D. Tank Modeling

To select the optimal propellant and pressurant tanks for the final configuration, and to ensure acceptable pressure and temperature evolution, a tank modeling code was developed. My primary focus during development was to design the correct sequence and respective conditions for calculating properties. We established boundary and initial conditions, incorporating necessary assumptions to ensure realistic predictions.

E. Piping&Instrumentation Diagram

The lines downstream of the propellant tanks were designed to prioritize the safety of ground crew from toxic propellants and to prevent unintended thrust caused by leakage during the mission. This is achieved using normally closed pyrovalves and double-seated solenoid valves, with additional latch valves for the main engine. Filters located after the pyrovalves are included to capture debris generated during valve activation. Additionally, test ports and fill and drain valves are incorporated to facilitate propellant filling, air drainage, and pressure testing.

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